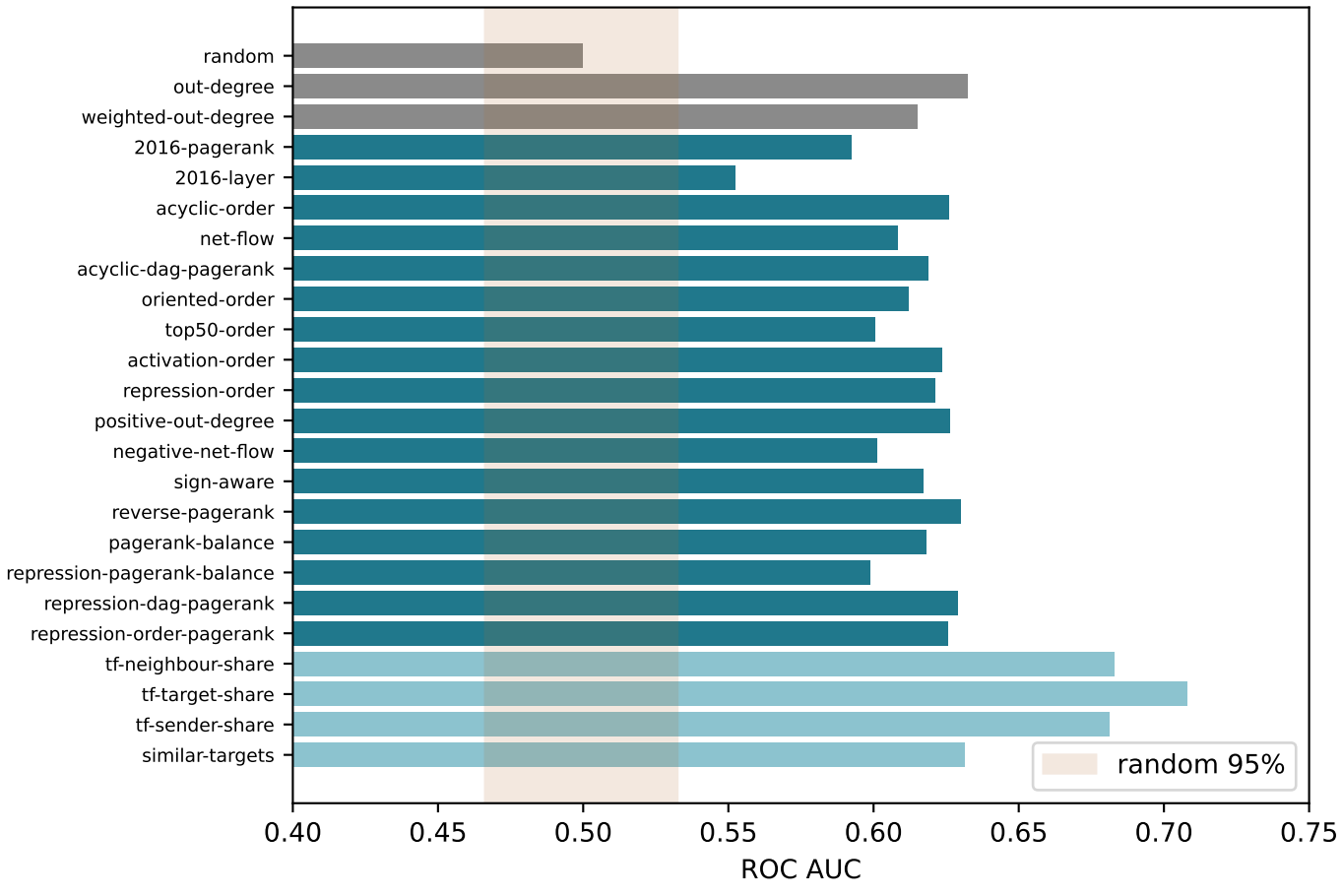
Regulator ranking against reference 'dream5-tfs'

2026-11-08 · 4,511 genes ranked, 334 in tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	7.3 (3–12)	0.500 (0.466–0.532)				
out-degree	12	0.632	0.055	< 0.001	+0.000 (+0.000, +0.000)	
weighted-out-degree	14	0.615	0.012	< 0.001	-0.018 (-0.024, -0.011)	< 0.001
2016-pagerank	3	0.592	0.981	< 0.001	-0.040 (-0.059, -0.019)	0.002
2016-layer	16	0.552	0.002	< 0.001	-0.080 (-0.108, -0.053)	< 0.001
acyclic-order	12	0.626	0.055	< 0.001	-0.006 (-0.015, +0.002)	0.168
net-flow	12	0.608	0.055	< 0.001	-0.024 (-0.036, -0.013)	< 0.001
acyclic-dag-pagerank	10	0.619	0.186	< 0.001	-0.014 (-0.022, -0.005)	0.002
oriented-order	19	0.612	< 0.001	< 0.001	-0.020 (-0.032, -0.009)	0.004
top50-order	18	0.600	< 0.001	< 0.001	-0.032 (-0.053, -0.012)	< 0.001
activation-order	14	0.624	0.012	< 0.001	-0.009 (-0.019, +0.001)	0.080
repression-order	14	0.621	0.012	< 0.001	-0.011 (-0.024, -0.000)	0.048
positive-out-degree †	12	0.626	0.055	< 0.001	-0.006 (-0.013, +0.000)	0.074
negative-net-flow †	14	0.601	0.012	< 0.001	-0.031 (-0.046, -0.017)	< 0.001
sign-aware †	13	0.617	0.025	< 0.001	-0.015 (-0.024, -0.007)	< 0.001
reverse-pagerank †	14	0.630	0.012	< 0.001	-0.003 (-0.008, +0.003)	0.354
pagerank-balance †	11	0.618	0.108	< 0.001	-0.014 (-0.029, +0.001)	0.066
repression-pagerank-balance †	12	0.599	0.055	< 0.001	-0.034 (-0.053, -0.014)	< 0.001
repression-dag-pagerank †	14	0.629	0.012	< 0.001	-0.003 (-0.013, +0.007)	0.530
repression-order-pagerank †	14	0.626	0.012	< 0.001	-0.007 (-0.018, +0.004)	0.234
tf-neighbour-share ‡	38	0.683	< 0.001	< 0.001	+0.050 (+0.004, +0.094)	0.028
tf-target-share ‡	32	0.708	< 0.001	< 0.001	+0.076 (+0.036, +0.113)	< 0.001
tf-sender-share ‡	34	0.681	< 0.001	< 0.001	+0.049 (+0.004, +0.089)	0.038
similar-targets ‡	31	0.631	< 0.001	< 0.001	-0.001 (-0.045, +0.038)	0.982

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).

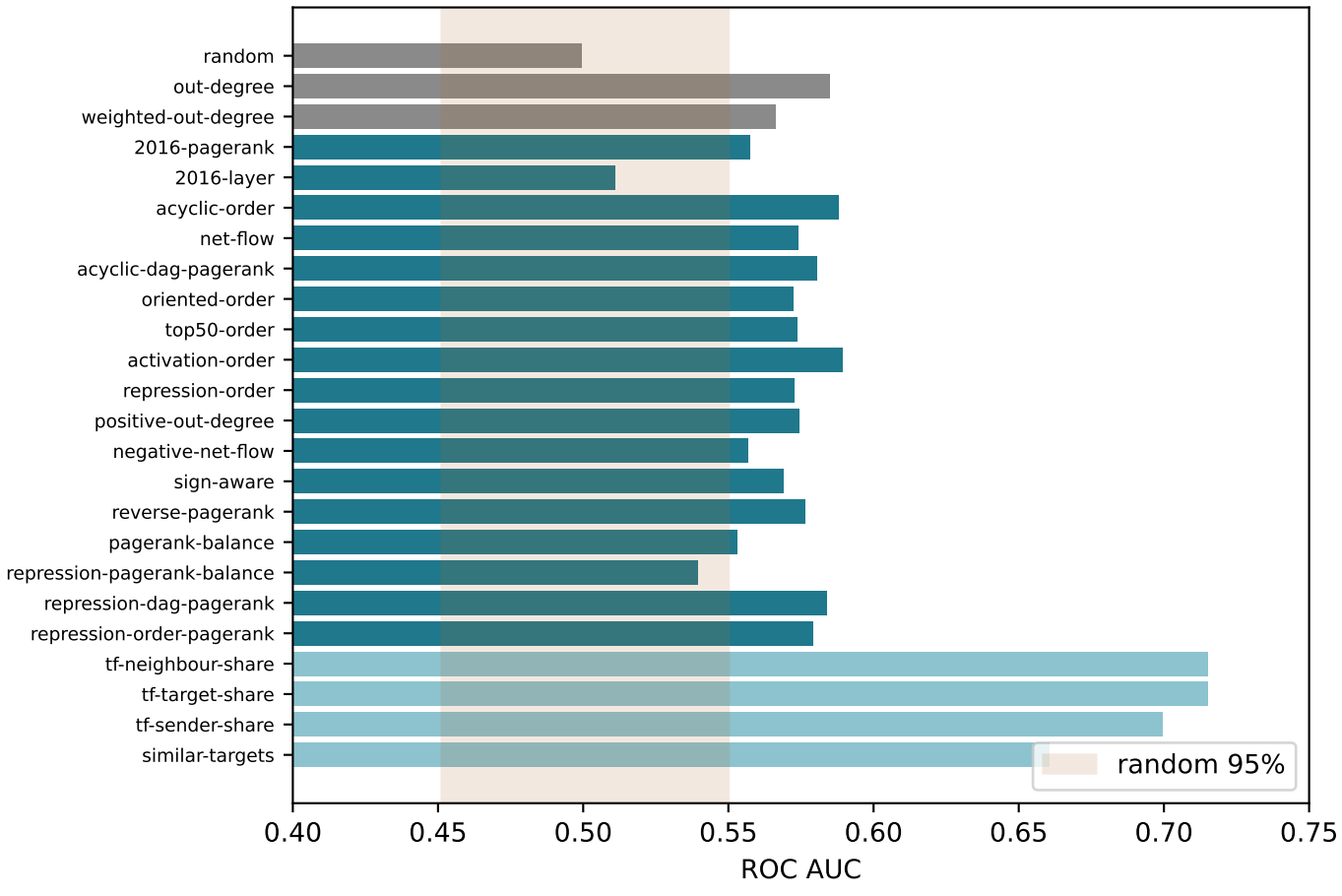


Regulator ranking against reference 'true-regulators'

2026-11-08 · 4,511 genes ranked, 141 in regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	3.1 (0–7)	0.499 (0.451–0.550)				
out-degree	0	0.585	1.000	< 0.001	+0.000 (+0.000, +0.000)	
weighted-out-degree	0	0.566	1.000	0.004	-0.019 (-0.028, -0.009)	< 0.001
2016-pagerank	1	0.557	0.958	0.013	-0.027 (-0.052, +0.002)	0.062
2016-layer	10	0.511	0.003	0.314	-0.074 (-0.114, -0.029)	< 0.001
acyclic-order	0	0.588	1.000	< 0.001	+0.003 (-0.010, +0.016)	0.634
net-flow	0	0.574	1.000	< 0.001	-0.011 (-0.026, +0.006)	0.178
acyclic-dag-pagerank	0	0.581	1.000	< 0.001	-0.004 (-0.017, +0.008)	0.472
oriented-order	0	0.572	1.000	0.003	-0.013 (-0.030, +0.006)	0.172
top50-order	0	0.574	1.000	< 0.001	-0.011 (-0.041, +0.019)	0.492
activation-order	0	0.589	1.000	< 0.001	+0.004 (-0.012, +0.020)	0.610
repression-order	0	0.573	1.000	0.003	-0.012 (-0.031, +0.006)	0.184
positive-out-degree †	0	0.574	1.000	< 0.001	-0.010 (-0.022, +0.001)	0.074
negative-net-flow †	0	0.557	1.000	0.014	-0.028 (-0.049, -0.006)	0.016
sign-aware †	0	0.569	1.000	0.004	-0.016 (-0.029, -0.003)	0.020
reverse-pagerank †	0	0.577	1.000	< 0.001	-0.008 (-0.016, -0.000)	0.038
pagerank-balance †	0	0.553	1.000	0.020	-0.032 (-0.053, -0.008)	0.006
repression-pagerank-balance †	0	0.540	1.000	0.056	-0.045 (-0.073, -0.018)	0.002
repression-dag-pagerank †	0	0.584	1.000	< 0.001	-0.001 (-0.016, +0.013)	0.874
repression-order-pagerank †	0	0.579	1.000	< 0.001	-0.006 (-0.022, +0.010)	0.478
tf-neighbour-share ‡	19	0.715	< 0.001	< 0.001	+0.130 (+0.062, +0.202)	< 0.001
tf-target-share ‡	16	0.715	< 0.001	< 0.001	+0.130 (+0.065, +0.197)	< 0.001
tf-sender-share ‡	17	0.700	< 0.001	< 0.001	+0.115 (+0.047, +0.189)	0.002
similar-targets ‡	15	0.660	< 0.001	< 0.001	+0.076 (+0.010, +0.146)	0.022

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

gold

Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	2,066	0.215 (8.93×)	12.59×	1.98×	
resolve_reciprocals → drop_sinks → prune_triangles	2,065	0.025 (2.91×)	4.64×	1.99×	
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	1,525	0.012 (2.45×)	3.33×	2.16×	
absolute	2,066	0.215 (8.93×)			
absolute → minimum_feedback_order	2,054	0.141 (7.66×)			
orient_reciprocals ratio=2.0 → absolute	2,037	0.022 (1.94×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	2,037	0.022 (1.94×)			
top_outgoing k=50 → absolute	2,066	0.182 (17.56×)			
top_outgoing k=50 → absolute → minimum_feedback_order	2,033	0.107 (15.61×)			
keep_sign sign=1 → absolute	2,066	0.199 (12.59×)			
keep_sign sign=1 → absolute → minimum_feedback_order	2,033	0.129 (10.86×)			
keep_sign sign=-1 → absolute	1,994	0.017 (1.98×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	1,795	0.015 (2.07×)			
keep_sign sign=-1	1,994	0.017 (1.98×)		1.98×	

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 516,484 edges (192,351 negative), largest strong component 4,511.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 516,484 edges (192,351 negative), largest strong component 4,511.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 124,935 edges (77,850 negative), largest strong component 2,621.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 124,935 edges (77,850 negative), largest strong component 2,621; DAG keeps 55,156 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 516,484 edges (0 negative), largest strong component 4,511; DAG keeps 390,471 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 516,484 edges (0 negative), largest strong component 4,511.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 516,484 edges (0 negative), largest strong component 4,511; DAG keeps 390,471 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 270,586 edges (0 negative), largest strong component 1; DAG keeps 270,586 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 199,033 edges (0 negative), largest strong component 4,511; DAG keeps 118,628 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 324,133 edges (0 negative), largest strong component 4,511; DAG keeps 231,141 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 192,351 edges (0 negative), largest strong component 4,126; DAG keeps 159,605 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 516,484 edges (192,351 negative), largest strong component 4,511.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 516,484 edges (192,351 negative), largest strong component 4,511.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 516,484 edges (192,351 negative), largest strong component 4,511.

reverse-pagerank: Highest reverse PageRank (raw graph)